

Inhibitor Investigation

Demo 14 of 16 · Part four, Example projects · about 8–10 minutes for all three stages

You need: index.html open in any modern browser, on a projector or on each student's phone. No network, no install, no accounts. Nothing is fetched at runtime, so it works with the wi-fi off.

Learning goals

- One per stage, in the learner's own words — these are the three Takeaway cards on screen, verbatim. If a student can say these three things and point at the evidence for each, the demo has done its job.
1. One exact rate can match more than one candidate curve at once — separating the patterns depends on which concentration you pick, not on how precisely you measure it.
 2. At $S = K_m$, this deliberately matched set makes competitive and uncompetitive both predict 25.0 exactly — evidence can be exact and still not discriminate every hypothesis.
 3. Any concentration other than the matched K_m tie separates these two candidates exactly in this idealized model; the model does not rank an informative choice by how large its predicted gap is.

Where this sits on the path

An AI-built classroom tool from Part four. Its goal is to show that exact evidence can still leave two causes tied, its engine is scripted (an idealized synthetic assay model), and its assumptions are one tap away in **Details**. It does not depend on any earlier demo, so it can be taught on its own; the habit it shares with the rest of Part four is naming the boundary between mechanism and claim.

One question for the session

Can one exact measurement always tell you which of several causes is at work?

"Every number this bench reports is exact — no noise, no fitting, no rounding. And you are still about to watch two different explanations survive the same exact number. Our job for the next few minutes is to find out why, and what it takes to actually break the tie."

Terms students see

Label	Meaning in this activity
Initial rate	How fast the enzyme converts substrate before conditions change; the one number every assay in this bench reports.
Vmax	The enzyme's fastest possible rate, fixed at 100 teaching units throughout.
Km	The substrate concentration where the uninhibited rate reaches half of Vmax; fixed at 2 teaching units. Every stage's "S = Km" language refers to this value.
a, b	The two factors that turn the baseline formula into a candidate inhibition pattern. Competitive: a=3, b=1. Uncompetitive: a=1, b=3. Pure noncompetitive: a=b=3.

Six substrate concentrations are offered throughout (0, 1, 2, 4, 8, 20 teaching units), chosen so the $S=K_m$ tie and its resolution are both reachable within the four-assay budget.

Stage 1 · Bench

Question: Does one exact assay point already tell you which inhibition pattern you are looking at?

Predict prompt: Before running anything, ask the room: can one assay at a single concentration ever be enough, by itself, to name the pattern? Take a show of hands.

Activity (3 minutes): 0:00 read the three candidate curves and the calibrated baseline on the chart · 0:45 choose a substrate concentration · 1:00 press **Run selected assay** · 1:30 compare the plotted point with all three curves · 2:30 try a second concentration if the budget allows.

Expected observation: at $S = 2$ (Km), the observed rate is exactly 25.0 whether the hidden pattern is competitive or uncompetitive — the point lands on both curves at once. At $S = 8$ or $S = 20$, the three curves are visibly separated and a single point discriminates one or more candidates.

Check-yourself answer key: "Yes — this deliberately matched set makes two curves cross exactly at $S = K_m$." The tempting wrong answer treats an exact number as automatically unique to one pattern; the chart's own crossing point at $S=2$ refutes it directly.

Teaching limit: do not let "the rate was exact" become "the rate was informative". Exactness rules out measurement noise; it does not by itself rule out other candidates. Stage 2 makes that gap explicit.

Misconception: an exact measurement cannot be ambiguous. It can, when two candidate formulas are built to agree at one input. Point at the $S=2$ point sitting on both the competitive and uncompetitive curves at once.

Stage 2 · Explain

Question: Given the assays collected in Stage 1, does the evidence isolate one pattern or leave more than one standing?

Predict prompt: With just one assay at $S = K_m$, can the evidence possibly isolate a single pattern? Ask for the reasoning, not just the answer.

Activity (2–3 minutes): 0:00 reread the evidence ledger from Stage 1 · 0:30 choose the explanation that follows from it · 1:00 read the revealed pattern name and the candidate-model card together.

Expected observation: with only the $S=2$ assay on the bench, "the measurements still leave more than one synthetic rate pattern possible" is the sound explanation; with a second, separating assay recorded, "the measurements isolate one synthetic rate pattern" becomes sound instead. The mystery pattern's name only appears in the feedback after a choice is made.

Check-yourself answer key: "No — recognizing insufficient evidence is itself a correct conclusion here." The wrong option treats a hedge as a failure to commit; the model itself marks "insufficient evidence" sound whenever it is true.

Teaching limit: this stage never claims to identify a real inhibitor's molecular binding site, whichever explanation a student picks. "The measurements identify a real inhibitor's molecular binding site" is offered as a wrong option specifically to be ruled out.

Misconception: a correct investigator always names a mechanism. Naming the actual limit of the evidence — "insufficient evidence" — is the correct scientific conclusion whenever two or more candidates remain compatible. This activity never scores a hedge as a wrong answer.

Stage 3 · Transfer

Question: Given the tie at $S = K_m$, which additional measurement would eliminate a candidate?

Predict prompt: Which of the four measurement options would separate competitive from uncompetitive? Ask students to commit before pressing anything.

Activity (2–3 minutes): 0:00 restate the $S=K_m$ tie from Stage 2 · 0:30 commit a prediction · 1:00 press a measurement option and read the actual candidate predictions and the eliminated-candidate count · 1:30 try a second option and compare.

Expected observation: $S = 2$ repeats the tie and eliminates 0 of 2 remaining candidates. $S = 1$, $S = 8$, and $S = 20$ each eliminate exactly 1 of 2, even though their predicted gaps are different sizes.

Check-yourself answer key: "No — any measurable difference eliminates a candidate here; the model does not rank by gap size." The wrong option imports a real-assay intuition (bigger gap, easier to trust) into a model that has no measurement error for a bigger gap to overcome.

Teaching limit: this is the point where the idealization is most likely to be over-generalized. The activity's own Details drawer for this stage says explicitly that a real, noisy assay would have good reason to prefer the larger gap — this model simply does not model that reason.

Misconception: a bigger predicted difference is always more informative. In a real, noisy assay, yes. In this exact, noiseless model, no — every measurable difference is equally decisive because there is no measurement error for a bigger gap to overcome. Both are true; they are claims about different models.

Model and boundaries

$$v = V_{\max} \cdot S / (a \cdot K_m + b \cdot S)$$

- **$V_{\max}$** = 100 rate units, **K_m** = 2 concentration units, fixed throughout.
- Baseline (no inhibitor): $a = b = 1$. Competitive: $a = 3$, $b = 1$. Uncompetitive: $a = 1$, $b = 3$. Pure noncompetitive: $a = b = 3$.
- Every observation is exact: there is no measurement error, curve-fitting, or hidden tolerance anywhere in the model.
- The four-assay budget is a teaching constraint chosen so the $S=K_m$ tie and its resolution are both reachable, not a real assay's practical limit.

This is a steady-state initial-rate teaching model only. It does not represent mixed, partial, allosteric, tight-binding, time-dependent, irreversible, or multiple-substrate inhibition behavior, and it never identifies a real inhibitor's molecular binding site or mechanism. Full per-stage assumption lists are in the app's **Details** drawer, which is the surface students should be sent to when they ask "but is that real?".

Discussion questions

- What claim is supported by the plotted points at any given moment? What claim would be too strong?
- Why does measuring the same concentration twice not add discrimination?
- What would a real assay need — replicates, controls, a broader concentration range, estimated error — before treating a fitted pattern as a real mechanism?
- The page labels some things Live and some Scripted. Pick one of each on screen and say what the label is promising you.
- In Stage 3, why does the model refuse to rank an informative measurement by the size of its predicted gap? When would that ranking matter?

Presenter controls

The three stage tabs run across the top and are never locked; "explain before reveal" is a within-stage content gate (the mystery pattern's name stays out of the Stage 2 feedback text until a choice is made), not a locked tab. **Details** in each stage intro opens that stage's formula, assumptions and sources. ? reopens the Guide. ⚙ Settings offers **Open Presenter Notes** (this document), **Presentation mode**, and **Reset**.

Presentation mode scales the page up one notch and adds a **Notes** button to the header for these cues. Reset returns the demo to its opening state *without reloading the page* — a fresh mystery sample, an empty bench, stage 1, and every explanation and transfer choice cleared — and *leaves presentation mode as you set it*. **New mystery sample** (in Stage 1) starts a fresh case without a full Reset, for running the bench a second time without restarting the whole demo.

Keep the label vocabulary honest. Every candidate curve and every recorded assay rate is labelled **Scripted** — a hand-built idealized formula, not a real assay reading. The live comparisons run on those numbers (whether an explanation is sound, how many candidates a measurement eliminates) are labelled **Live**, because they are computed on the device the moment a student acts, even though the numbers they compare are Scripted.

Sources and attribution

- Strelow J, Dewe W, Iversen PW, et al. [Mechanism of Action Assays for Enzymes](#), NIH Assay Guidance Manual — idealized initial-rate inhibition-pattern context and the value of measurements above K_m , cited throughout the app's Details drawer.

This source supports the underlying assay-design context; it does not endorse this activity. A Murray State biochemistry course served as the reviewed course-fit reference; it is not named here, and it does not supply current scheduling information.

Inhibitor Investigation · Bryant Harrison · Murray State University · Offline teaching model. Open `index.html` to run the investigation. `src/demo-guide.html` is the canonical source of this document; the shipped `teacher-guide.html`, the PDF beside it, and the in-app presenter notes are all generated from it at build time.